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Coulombkraft:

$$\vec{F}_C = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{|\vec{r}|^2}\cdot\frac{\vec{r}}{|\vec{r}|}$$

$$|\vec{F}_C| = \frac{q_1 q_2}{4\pi\varepsilon_0 r^2}$$

Potential:

$$F_C = -\frac{\partial E_{pot}}{\partial r}$$

$$E_{pot}(r) = \frac{q_1 q_2}{4\pi\varepsilon_0 r}$$

$U(r) :=$ el. Potential [V]

$$U(r) = \frac{E_{pot}(r)}{q_2} = \frac{q_1}{4\pi\varepsilon_0 r}$$

mit n-Punktladungen:

$$U_{P,ges} = \sum_{i=1}^{N} u_i = \frac{1}{4\pi\varepsilon_0}\sum_{i=1}^{N}\frac{q_i}{r_i}$$

$$U_{pp}(x) = U_{max}\frac{x}{L}$$

$$\Delta E_{pot} = q U_{max} = q\,\Delta U$$

$$E_{pot}(x) = -e\,U_{pp}(x) = -e\,U_{max}\frac{x}{L} = -e\,U\frac{x}{L}$$

$$F_x(x) = -\frac{\partial}{\partial x}E_{pot}(x) = \frac{eU}{L}$$

$$a_x = \frac{F_x}{m_e} = \frac{eU}{m_e L} \qquad v_x = \sqrt{2a_x L} = \sqrt{\frac{2eU}{m_e}}$$

Ablenkung vom Kondensator

$$\vec{v} = \begin{pmatrix} a_x t \\ v_y \end{pmatrix} = \begin{pmatrix} \frac{e U_{max}}{m_e e} \\ v_y \end{pmatrix}$$

Das Elektrische Feld

$$\vec{F}_C(\vec{r}) = q_2\vec{E}_C(\vec{r}), \quad \vec{E}_C(\vec{r}) = \frac{1}{4\pi\varepsilon_0}\frac{q_1}{|\vec{r}|^2}\frac{\vec{r}}{|\vec{r}|}$$

$$|\vec{E}| = \frac{U}{L} \quad \vec{F} = q\vec{E}, \quad F_x(x) = -e|\vec{E}|$$

(Kondensator)

Stokessche Reibung:

$$\vec{F}_S = -b_S|\vec{r}|\frac{\vec{v}}{|\vec{v}|}$$

Temperaturabhängigkeit des spez. Widerstands:

$$\rho(T) = \rho_0(1+\alpha(T-T_0)) \quad \alpha = \text{Temp. Koeffizient}$$

Spannungsquellen $U_B = \frac{\partial W}{\partial Q}$ $\quad P_{el} = \frac{\partial W}{\partial t} = \frac{\partial W}{\partial Q}\frac{\partial Q}{\partial t} = U_B I = \frac{U_B^2}{R}$

$U_{B,ges} = \sum_{i=1}^{N} U_B$, unverzweigte Gleichstromkreise $U_B - U_R = U_B - IR = 0$

$U_B = RI$

Batterie mit Innen-R: $U_B - IR_{innen} - IR = 0$ $\quad U_{Klemm} = U_B - IR_{innen}$

R in Reihe: $U - I\sum_{i=1}^{N} R_i = 0$

Potential / Spannungsdifferenz im Stromkreis $U_1 - U_2 = IR$ auch Spannungsabfall an R

Verzweigte Stromkreise, Parallelschaltung von Widerständen:

$U = I_1 R_1 = I_2 R_2 = I_3 R_3 \Rightarrow$ Kirchhoff

$I = I_1 + I_2 + I_3 = U\left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}\right)$ $\quad R_{eq} = \left(\sum_{i=1}^{N}\frac{1}{R_i}\right)^{-1}$ $\quad U = IR_{eq}$

Kondensatoren $|\vec{E}| = \frac{U}{L}$ $\quad U = U_B$ $\quad Q = CU$ $\quad C = \varepsilon_0\frac{A}{d}$

Energie $E_C = \frac{1}{2}CU_0^2$, parallel: $U = U_B$ $\quad q_1 = C_1 U$, $q_2 = C_2 U$,

Reihe: $U_1 = \frac{q}{C_1}$, $U = U_1 + U_2 \cdots = q\left(\frac{1}{C_1} + \frac{1}{C_2}\ldots\right)$, $C_{eq} = \frac{Q}{U}$, $Q = \sum_{i=1}^{N} q_i$, $C_{eq} = \sum_{i=1}^{N} C_i$

$U = \frac{Q}{C_{eq}}$, C_{eq} := Gesamtkapazität $= \left(\sum_{i=1}^{N}\frac{1}{C_i}\right)^{-1}$

Der RC-Kreis $U_0 - IR - \frac{Q}{C} = 0$ $\quad Q_0 = CU_0$ $\quad I = \frac{\partial Q}{\partial t}$

Entladung: $Q(t) = Q_0\exp\left(-\frac{t}{\tau_C}\right)$ $\quad \tau_C = RC$ $\quad Q_{\tau_C} = \frac{Q_0}{e}$ $\quad t_{C,\frac{1}{2}} = \ln(2)\tau_C$

$U(t) = U_0\exp\left(-\frac{t}{\tau_C}\right)$, $I(t) = -I_0\exp\left(-\frac{t}{\tau_C}\right)$, $I_0 = \frac{U_0}{R}$

Aufladung wenn $Q(0) = 0$: $Q(t) = CU_0\left(1 - \exp\left(-\frac{t}{RC}\right)\right) = Q_0\left(1 - \exp\left(-\frac{t}{\tau_C}\right)\right)$

$U_C(t) = \frac{Q_0}{C}\left(1 - \exp\left(-\frac{t}{\tau_C}\right)\right) = U_0\left(1 - \exp\left(-\frac{t}{\tau_C}\right)\right)$, $I(t) = I_C e^{-t/\tau_C}$

Induktivität $U = -L\frac{\partial I}{\partial t} = -U_2$ $\quad L = \mu_0\frac{N^2 A}{L}$

R-L-Kreis: $U_0 - IR - L\frac{\partial I}{\partial t} = 0$

Einschalten der Spule: $I(t) = \frac{U}{R}\left(1 - \exp\left(-\frac{R}{L}t\right)\right) = I_0\left(1 - e^{-t/\tau_L}\right)$

Zeitkonstante: $\tau_L = \frac{L}{R}$ $\quad I_0 = \frac{U_0}{R}$

Spannungsabfall an Spule: $U_R(t) = U_0\left(1 - \exp\left(-\frac{R}{L}t\right)\right) = U_0\left(1 - e^{-t/\tau_C}\right)$

Induktionsspannung: $U_L(t) = U_0 - U_R(t) = U_0 e^{-t/\tau_C}$

Ausschalten: $I = \frac{U_0}{R}e^{-t/\tau_C} = I_0 e^{-t/\tau_L}$, $U_L(t) = -U_0 e^{-t/\tau_2}$

Elektrischer Strom

$$I = \frac{\partial Q}{\partial t} \qquad Q = \int_{t_1}^{t_2} I\, dt$$

$$I = \frac{-eU}{\Delta x} = \frac{qU}{\Delta x}$$

Ladungsträgerdichte → $n = \frac{N}{V}$ ← freie Ladungsträger, ← Volumen

$\rho_q = nq = \frac{N}{V} q$:= Ladungsdichte

$n_{el} = \frac{Q}{C}$

$I = n_{el}(-e) v \cdot A = jA$

$j := n_{el}(-e) v$:= Stromdichte $[\frac{A}{m^2}]$

$|\vec{F}_{el}| = \frac{eU}{L}$

Driftgeschw. $= V_{max} = V_D$

$$v_D = -\frac{I}{n_{el} e A} = -\frac{j}{n_{el} e}$$

$$I = \frac{U}{R}$$

Elektrischer Strom u. E-Erhaltung:

$\Delta E_{pot} = -e(U_2 - U_1) = -eU$

$\Delta E_{kin} = E_{kin,2} - E_{kin,1} = \frac{1}{2} m v_0^2 - \frac{1}{2} m v_0^2 = 0$

$$\frac{\partial(\Delta E_{pot,ges})}{\partial t} = \frac{\partial Q}{\partial t} U = IU$$

El. Leistung $P_{el} = U \cdot I$ Einheit $[W] = [js]$

Ohmscher Widerstand:

$$R = \frac{U}{I} \qquad P = UI = I^2 R = \frac{U^2}{R}$$

Der Spezifische Widerstand $[\frac{\Omega mm^2}{m}]$

$R = \rho \frac{L}{A} \Rightarrow \rho = R \frac{A}{L}$ $[\Omega m]$ $\quad \sigma = \frac{1}{\rho}$ spez. Leitfähigkeit

$\vec{E} = \rho \vec{j}$

$$\rho = \frac{|\vec{E}|}{|\vec{j}|} = \frac{U/L}{I/A} = R\frac{A}{L}$$

Drude-Modell:

$a = \frac{F_{el}}{m_e} = \frac{e|\vec{E}|}{m_e}$, $|v_D| = a\tau = \frac{e|\vec{E}|\tau}{m_e}$ $\quad \rho = \frac{m_e}{e^2 n_{el} \tau}$

τ Zeit vor Stoß $\quad |v_D| = \frac{|\vec{j}|}{n_{el} e} \quad |\vec{E}| = \left(\frac{m_e}{e^2 n_{el} \tau}\right) |\vec{j}|$

Selbstinduktion, Ausschalten:

$$I(t) = I_0 e^{-t/\tau_L} = \frac{U}{R_1} e^{-\frac{R_2}{L} t}$$

$$U_{L,0} = U \frac{R_2}{R_1} \qquad \tau_L = \frac{L}{R_2} \qquad U_L(t) = -U \frac{R_2}{R_1} e^{-\frac{R_2}{L} t}$$

Spule als E-Speicher: $E_{pot} = \frac{1}{2} R I_0^2 \frac{L}{R} = \frac{1}{2} L I_0^2$

Wechselstrom:

$U(t) = U_0 \cos(\omega t) = U_0 \cos(f 2\pi t)$

$I(t) = I_0 \cos(\omega t + \varphi) = I_0 \cos(2\pi f t + \varphi)$

$\tilde{U}(t) = U_0 e^{i(\omega t)}, \quad \tilde{I}(t) = I_0 e^{i(\omega t + \varphi)}$

R in Wechselstrom:

$$I_R(t) = \frac{U(t)}{R} = \frac{U_0}{R}\cos(\omega t) = I_0 \cos(\omega t) \qquad I_0 = \frac{U_0}{R} = \frac{U_0}{|Z_R|}$$

C in Wechselstrom:

$U(t) = U_0 \cos(\omega t) \quad Q(t) = C U_0 \cos(\omega t) \quad I_C = -C U_0 \omega \sin(\omega t) = I_0 \cos(\omega t + \frac{\pi}{2})$

$I_0 = U_0 \omega C = \frac{U_0}{|Z_C|} \quad |Z_C| = \frac{1}{\omega C}$

L in Wechselstrom:

$$\frac{\partial I_L}{\partial t} = \frac{U_0}{L}\cos(\omega t) \qquad I_L(t) = I_0 \sin(\omega t) = I_0 \cos(\omega t - \frac{\pi}{2})$$

$$I_0 = \frac{U_0}{\omega L} = \frac{U_0}{|Z_L|} \qquad |Z_L| = \omega L$$

Impedanz

$\tilde{U}(t) = U_0 e^{i(\omega t)}, \quad \tilde{I}(t) = i\omega C \tilde{U}(t)$

$Z_C = \frac{1}{i\omega C} \quad Z_C = |Z_C| e^{-i\varphi_C} \quad \varphi_C = \frac{\pi}{2} \quad \tilde{I}(t) = I_0 e^{i(\omega t + \varphi_C)}$

$Z_L = i\omega L = |Z_L| e^{-i\varphi_L} \quad \varphi_L = -\frac{\pi}{2} \quad Re(\tilde{I}(t)) = I_0 \cos(\omega t + \varphi_C)$

$Z_R = R = |Z_R| e^{-i \cdot 0} = |Z_R| e^{-\varphi_R} \quad \varphi_R = 0$

$Z_{eq} = \sum_{i=1}^{N} Z_i$ (Reihe), $\quad Z_{eq} = \left(\sum_{i=1}^{N} \frac{1}{Z_i}\right)^{-1}$ (Parallel)

Kombinationen im Wechselstromkreis:

ohmscher Spannungsteiler $\tilde{U}(t) = U_0 e^{i\omega t}$

$$\tilde{I}(t) = \frac{U_0}{R_{eq}} e^{i\omega t} = \frac{U_0}{R_1 + R_2} e^{i\omega t}$$

$$\tilde{U}_1(t) = \tilde{I}(t) R_1 = \frac{R_1}{R_1 + R_2} U_0 e^{i\omega t} = \frac{R_1}{R_1 + R_2} \tilde{U}(t)$$

$$U_2 = \frac{R_2}{R_{eq}} U_0 \qquad R_{eq} = Z_{eq} = R_1 + R_2$$

$$\int \rho\, dV = \varepsilon_0 \oint \vec{E} \cdot d\vec{A} = \varepsilon_0 \int \vec{\nabla} \cdot \vec{E}\, dV$$

$$\nabla \cdot \vec{E} = \frac{\rho}{\varepsilon_0}$$